8. Secular Stagnation

Adv. Macro: Heterogeneous Agent Models

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Introduction

Secular Stagnation

- **Today:** Can we explain *secular stagnation* through the lens of heterogeneous agent models?

- **Central economic questions:**
  1. Why have real interest rates fallen since the 1950s?
  2. What role did aging and rising inequality play?
  3. What happens next, and what does it mean for public debt?

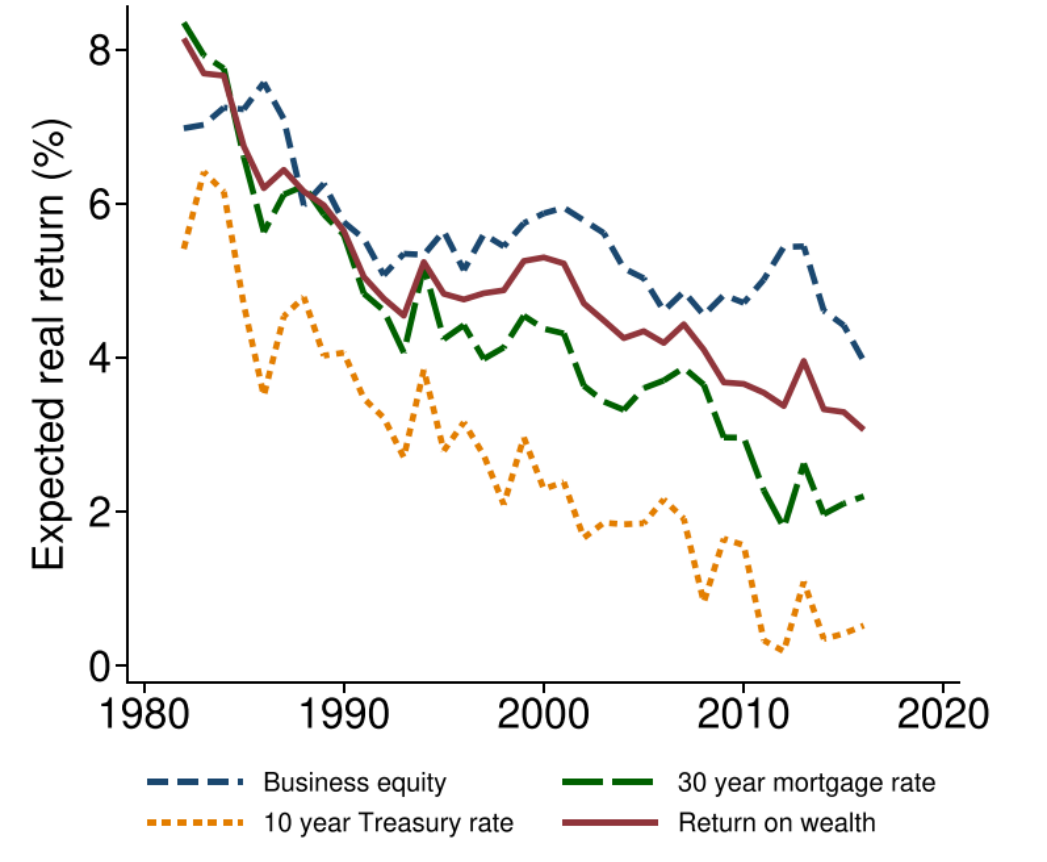
- **Plan for today:**
  1. Income inequality and saving (Straub 2019)
  2. The race between asset supply and asset demand (Auclert, Malmberg, Rognlie and Straub 2025)
  3. Indebted demand (Mian, Straub and Sufi 2021)

Secular Stagnation

- **Secular stagnation**: A state in which private demand is structurally low
  - Low level of growth in the economy
  - Low interest rates
  - Low level of inflation
- Large literature suggests that advanced economies have been in this state over the past $\approx$ 20 years

Declining interest rates

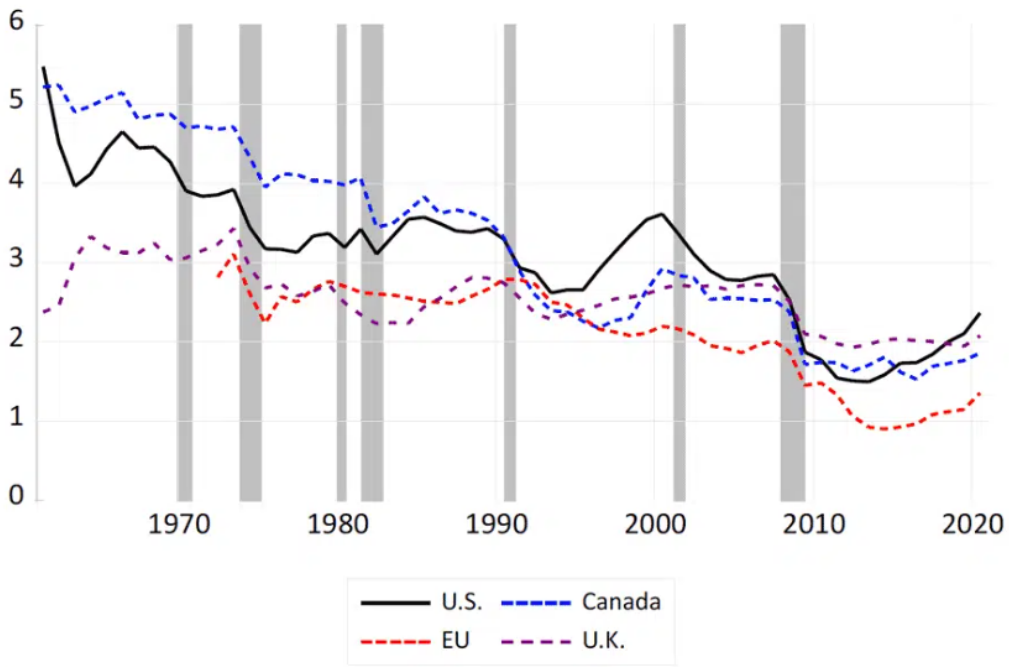
Various interest rates from Mian, Straub and Sufi (2021)



Declining growth

Trend Real Output Growth, 1961 to 2020

Percent



Source: www.cbo.gov/system/files/2020-12/56891-real%20-interest-rates.pdf

Data sources: Holston et al. (2017) and Federal Reserve Bank of New York, "Measuring the Natural Rate of Interest" (accessed December 10, 2020), www.newyorkfed.org/research/policy/rstar.

Rising saving rates and declining r

Mankiw (2022) writes a simple Solow model to think about secular stagnation

- Recall that, on the BGP $r = \alpha \frac{n+g+\delta}{s} - \delta$
- An increase in savings implies $dr/ds = -\alpha \frac{n+g+\delta}{s^2} < 0$
- Assume $\alpha = 1/3$, $n = 0.01$, $g = 0.02$, $\delta = 0.05$, $s = 0.24$, this yields $dr/ds = -0.46$ p.p.
- But population and GDP growth have declined as well! $dr/d(n+g) = 1.39$ p.p.

Other potential culprit: rising markups.
Implications for fiscal and monetary policy? Low r means more likely to hit ZLB and fiscal stimulus cheap and powerful.

Explanations

Large literature on potential explanations

- Increases in market power of firms
  - Liu, Mian & Sufi (2022), Aghion, Bergeaud, Boppart, Klenow & Li (2023)
- Declining relative price of investment (/reduced innovation)
  - Eichengreen (2015), Eggertsson, Mehrotra & Robbins (2019)
- **Aging population (Demographics)**
  - Auclert, Malmberg, Martenet & Rognlie (2024), Gagnon, Johannsen & López-Salido (2021)
- **Increasing income inequality**
  - Straub (2019), Mian, Straub & Sufi (2021), Auclert & Rognlie (2020)
- All of these in one framework: Auclert, Malmberg, Rognlie & Straub (2025)

Inequality and saving
Straub (2019)

Secular stagnation and income inequality

- A literature explains the decline in r through *rising* income/wealth **inequality**
  - See e.g. Mian, Straub and Sufi (2021): *What explains the decline in r*? Rising income inequality versus demographic shifts*
- **Example**: If saving rates increase with income (richer save more) then:
  - Redistribution from poor to rich households *increase* the aggregate supply of savings $\Rightarrow$ lower rates in GE
  - Suggests link between increasing income inequality and secular stagnation
  - Straub (2019, WP) tests this hypothesis

Stylized model

- Deterministic model with one-period lived households
- Constant wage w, return R
- Get utility from consumption c_t and wealth a_{t+1}

$$\max_{c_t, a_{t+1}} u(c_t) + \beta U(a_{t+1})$$

$$c_t + R^{-1} a_{t+1} \leq a_t + w_t$$

with $u(c) = \frac{c^{1-\sigma}}{1-\sigma}$ and $U(a) = \frac{a^{1-\Sigma}}{1-\Sigma}$
We can show that $c_t \approx k(a_t + w)^{\phi}$, with $\phi = \frac{\Sigma}{\sigma}$

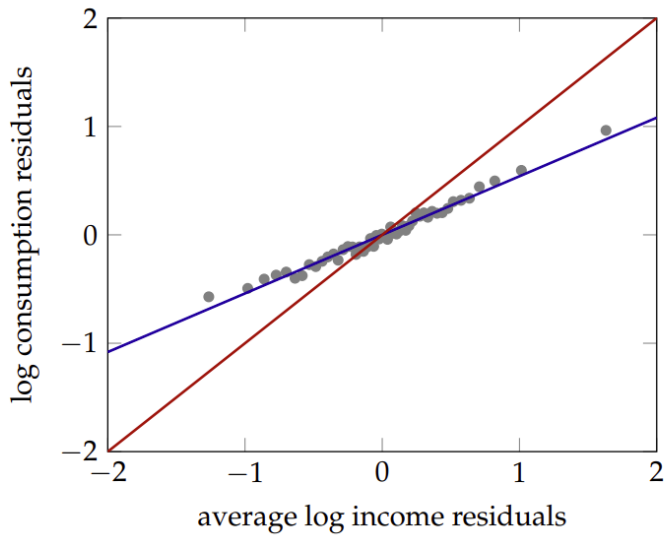
Empirical estimate

- We can check in the data the empirical relationship between permanent income and consumption
- If $\phi < 1$ consumption rises *less* than proportionally with income
  - I.e. savings rise *more* $\Rightarrow$ Richer HHs save more
  - Opposite for $\phi > 1$
- Two questions
  - What does this elasticity look like empirically?
  - If $\phi \neq 1$ how can we accommodate this in a HA model?

Empirical estimate

- Use US panel survey data (PSID)
- Regress log c_{it} on controls (year, age, HH size, location) and 9 year average of income
- Binned scatter:

Figure 3: Consumption and average income.



- More detailed estimates in paper, suggest $\phi \approx 0.7$

Permanent redistribution in the canonical HA model

- Consider standard HA model with permanent income state p :

$$V(a_{t-1}, z_t, p) = \max_{c_t} \frac{c_t^{1-\sigma}}{1-\sigma} + \beta \mathbb{E}\left[V(a_t, z_{t+1}, p)\right]$$

$$\text{subject to}$$

$$c_t + a_t = R a_{t-1} + z_t p$$

$$a_t \geq 0$$

$$\ln z_t = \rho \ln z_{t-1} + \epsilon_t$$

- Note: preferences are *homothetic* (homogeneous of degree 1)
  - »Scale independent«

Homothetic household problem I

- Normalize constraints by p and Bellman by $p^{1-\sigma}$

$$\frac{V\left(a_{t-1}, z_t, p\right)}{p^{1-\sigma}} = \max_{c_t} \frac{\frac{c_t^{1-\sigma}}{1-\sigma}}{p^{1-\sigma}} + \beta \frac{\mathbb{E}\left[V\left(a_t, z_{t+1}, p\right)\right]}{p^{1-\sigma}}$$

subject to

$$\frac{c_t}{p} + \frac{a_t}{p} = R\frac{a_{t-1}}{p} + z_t\frac{p}{p}$$

$$\frac{a_t}{p} \geq \frac{0}{p}$$

Homothetic household problem II

Define $\tilde{c}_t = \frac{c_t}{p}, \tilde{a}_t = \frac{a_t}{p}, \tilde{V}_t = \frac{V_t}{p^{1-\sigma}}$ to get:

$$\tilde{V}\left(\tilde{a}_{t-1}, z_t\right) = \max_{\tilde{c}_t} \frac{\tilde{c}_t^{1-\sigma}}{1-\sigma} + \beta \mathbb{E}\left[\tilde{V}\left(\tilde{a}_t, z_{t+1}\right)\right]$$

$$\text{subject to}$$

$$\tilde{c}_t + \tilde{a}_t = R\tilde{a}_{t-1} + z_t$$

$$\tilde{a}_t \geq 0$$

$$\ln z_t = \rho \ln z_{t-1} + \epsilon_t$$

Then HH problem is independent of p up to scale!

- Solution is "scale independent": 1% more permanent income raises c_t and a_t by 1%
- Homothetic preferences imply $\phi = 1$, since $\tilde{c}_t = c_t/p$
- Permanent redistribution has *no effect* on aggregates: the lower saving of one group exactly offsets the higher saving of the other

Non-homothetic HA model

- Extend standard HA model with *taste for wealth* (»status«)

$$V(a_{t-1}, z_t, p) = \max_{c_t} \frac{c_t^{1-\sigma}}{1-\sigma} + \phi_a \frac{a_t^{1-\sigma_a}}{1-\sigma_a} + \beta \mathbb{E}\left[V(a_t, z_{t+1}, p)\right]$$

$$\text{subject to}$$

$$c_t + a_t = a_{t-1}(1+r) + z_t p$$

$$a_t \geq 0$$

$$\ln z_t = \rho \ln z_{t-1} + \epsilon_t$$

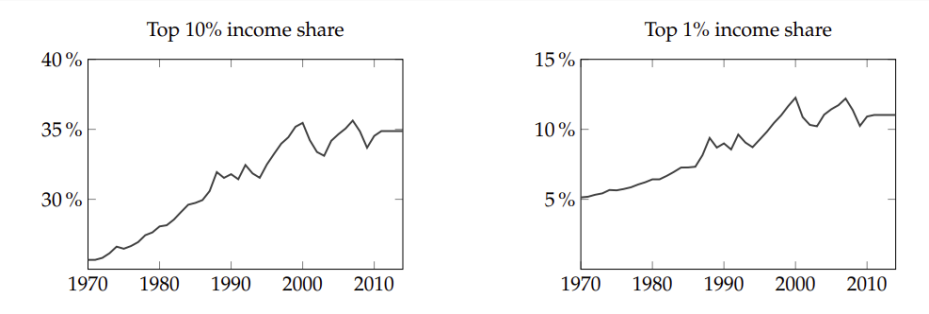
- Note: Still homothetic as long as $\sigma = \sigma_a$, but non-homothetic if $\sigma \neq \sigma_a$:

$$\tilde{V}(\tilde{a}_{t-1}, z_t, p) = \frac{\tilde{c}_t^{1-\sigma}}{1-\sigma} + p^{\sigma-\sigma_a}\phi_a \frac{\tilde{a}_t^{1-\sigma_a}}{1-\sigma_a} + \beta \mathbb{E}\left[\tilde{V}(\tilde{a}_t, z_{t+1}, p)\right]$$

- If $\sigma \neq \sigma_a$, the weight $p^{\sigma-\sigma_a}$ on wealth depends on p: richer households put more weight on wealth when $\sigma > \sigma_a$

Applications

- Calibrate a life-cycle HA model with non-homothetic preferences to estimate $\phi = 0.7$
  - Note: In paper there is a second source of non-homotheticity where σ varies across age
- Segment HHs into 3 permanent income groups corresponding to poorest 90%, next 9% and richest 1%
- Feed in evolution of income inequality from Piketty and Saez (2003), keeping aggregate income unchanged in PE (so permanent redistribution between rich and poor HHs)

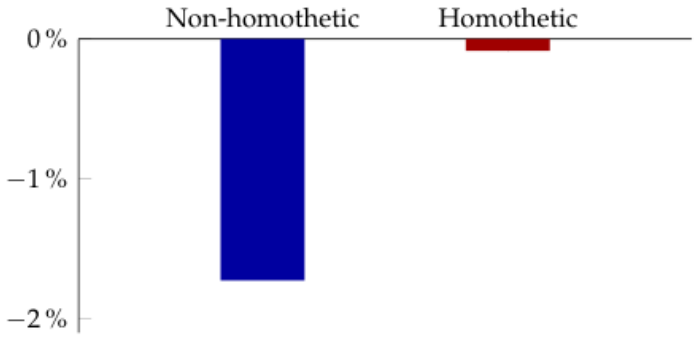


- Solve for GE (Supply side as in HANC)

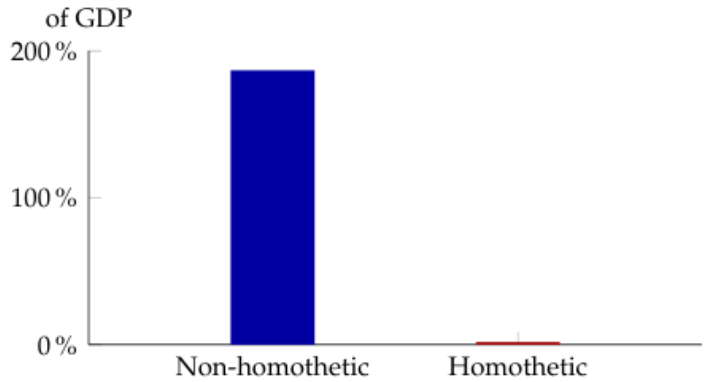
Results

Figure 6: The partial equilibrium effects of a rising income inequality.

(a) Short-run drop in aggregate consumption.

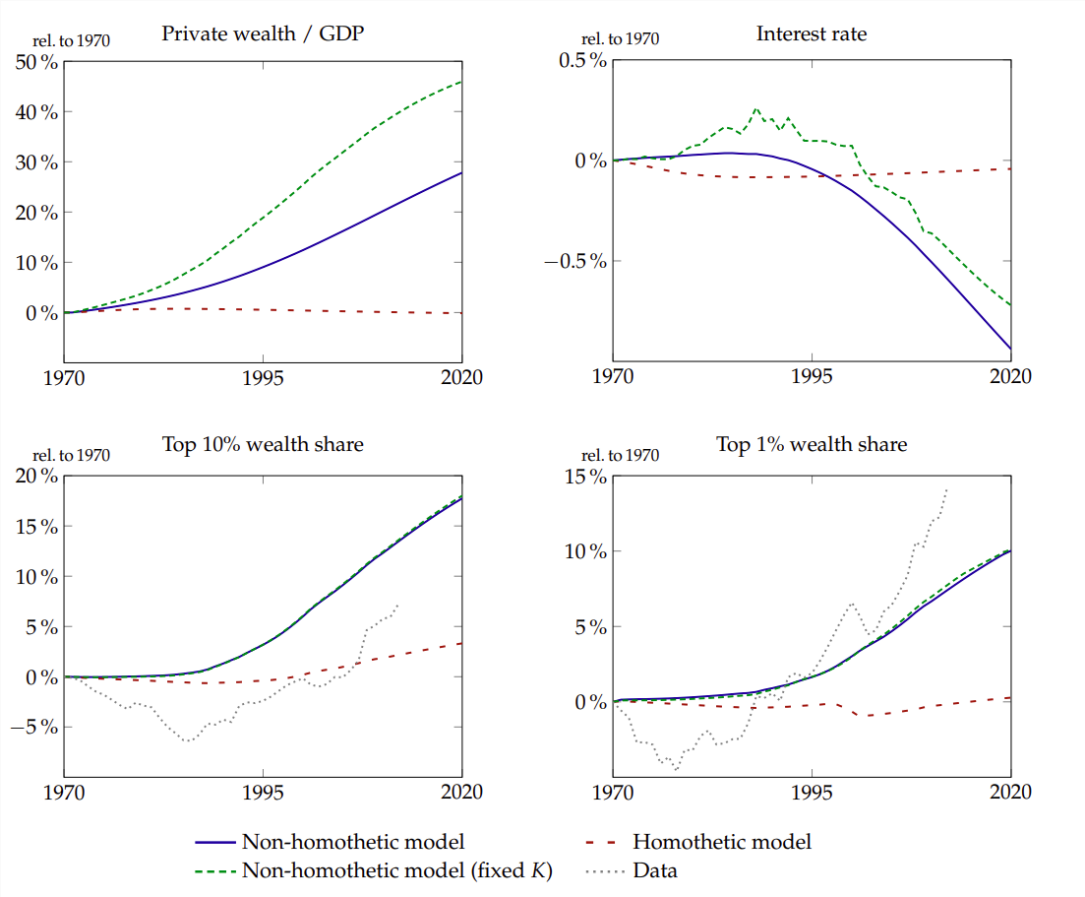


(b) Long-run rise in aggregate savings.



GE implications of rising income inequality

- Compare effects in homothetic and non-homothetic models:



- Note: »Fixed K« is Lucas-tree economy, where wealth grows due to rising asset prices

Trickle down economics?

- In model with perfect competition: $\uparrow A \rightarrow \uparrow K \rightarrow \uparrow Y, w$
- The GE effect benefits to everybody!
- How realistic is this? Need to account for imperfect competition
- In this case: $r_t = \frac{\alpha}{\mu} K_t^{(\alpha-1)} - \delta$, and $q_t = \frac{p_{t+1}+q_{t+1}}{1+r_t}$
- Increase in savings might not finance more physical capital, but just higher prices of financial assets!

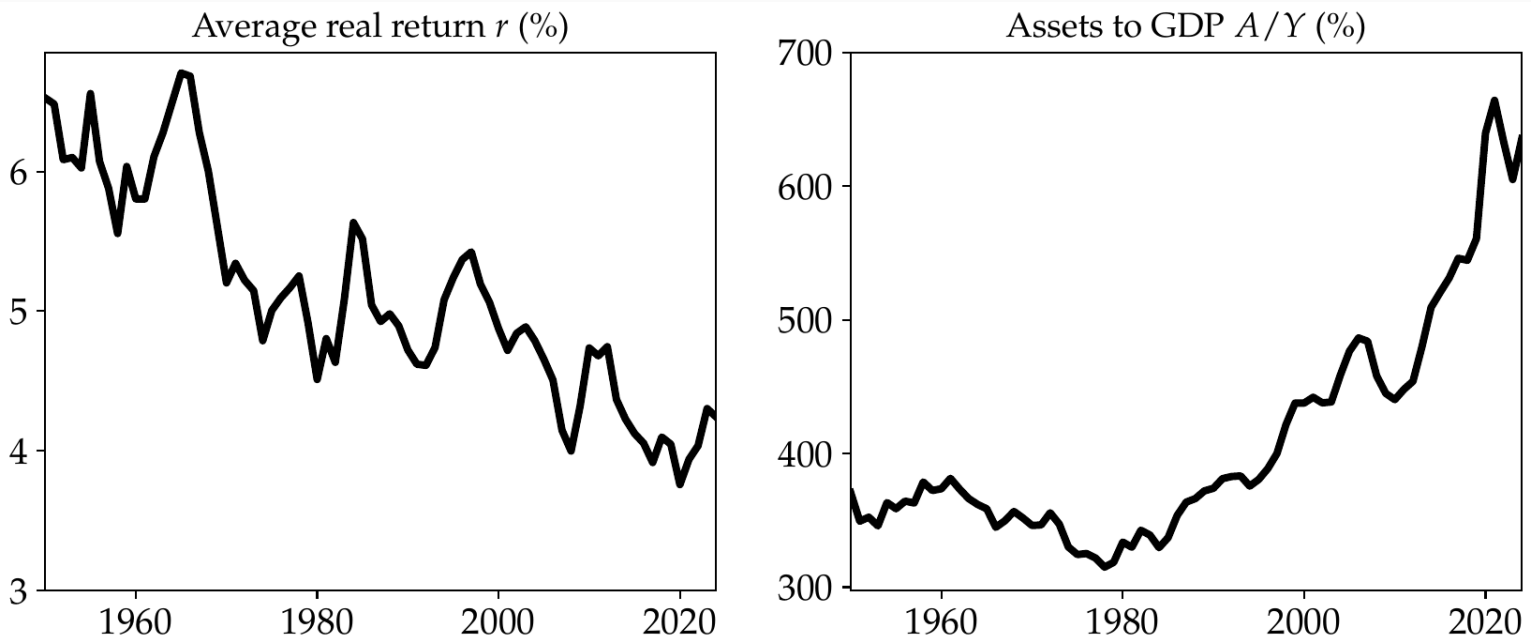
Asset supply and demand

Auclert, Malmberg, Rognlie and Straub (2025)

The race between asset supply and asset demand

- Auclert, Malmberg, Rognlie and Straub (2025), AMRS below. The U.S. since 1950:
  - fertility fell from over 3 children per woman to 1.6: the population ages
  - income inequality has risen sharply since the 1970s
  - government debt and the value of private assets rose, net foreign assets at a record low
- Two questions:
  1. How did these shifts affect interest rates and wealth over the last 75 years?
  2. What happens over the next 75 years, and what does it mean for public debt?
- Method: long-run asset supply and demand, measured with *sufficient statistics* from a life-cycle model
  - extends Auclert, Malmberg, Martenet and Rognlie (2024), who study demographics only

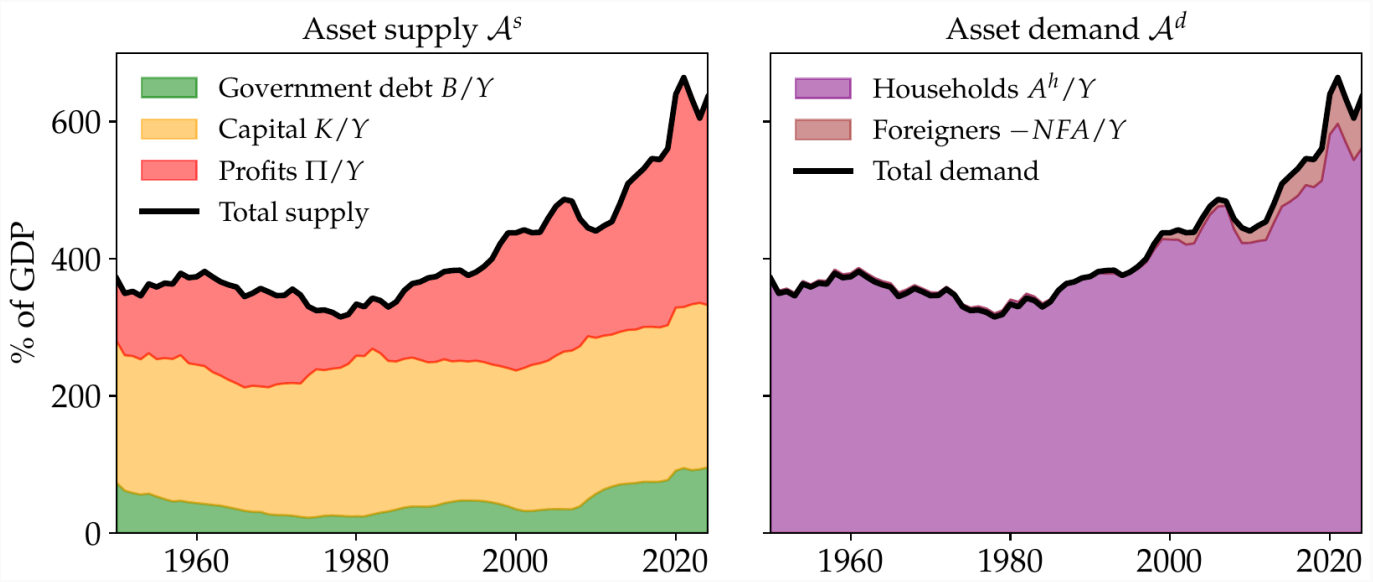
Falling returns and rising wealth



- Average return $r = \bar{s}r^f + (1 - \bar{s})r^r$: safe rate on government bonds r^f and return on private assets r^r, with $\bar{s} = 0.11$
- Since the 1950s, r fell by about 2 pp and wealth rose by about 300% of GDP, mostly after 1980

Source: AMRS (2025), Figure 2

Realized asset supply and demand, 1950-2024



- Supply: government debt B, capital K, capitalized profits and rents Π. Demand: households A^h and foreigners $-NFA$
- B, K, Y from the national accounts, A^h and NFA from the financial accounts, Π as the residual
- Supply rose through government debt and profits, demand through households and foreigners

Source: AMRS (2025), Figure 3

Asset market equilibrium

- The long-run rate r^* clears the market for assets, relative to GDP:

$$\mathcal{A}^s(r^*) = \mathcal{A}^d(r^*)$$

- **Asset supply**: domestic assets available to hold, decreasing in r (less capital, profits worth less)

$$\mathcal{A}^s = \underbrace{\frac{B}{Y}}_{\text{gov. debt}} + \underbrace{\frac{K}{Y}}_{\text{capital}} + \underbrace{\frac{\Pi}{Y}}_{\text{profits and rents}}$$

- **Asset demand**: assets that households and foreigners want to hold, increasing in r

$$\mathcal{A}^d = \underbrace{\frac{A^h}{Y}}_{\text{households}} + \underbrace{\left(-\frac{NFA}{Y}\right)}_{\text{foreigners}}$$

- Vocabulary: household saving is a *demand* for assets. Same diagram as Aiyagari (lectures 3-4), with B and Π added

Measuring the shifts

- Let η shift both curves, and define the semi-elasticities

$$\epsilon^s \equiv -\frac{\partial \log \mathcal{A}^s}{\partial r} > 0, \qquad \epsilon^d \equiv \frac{\partial \log \mathcal{A}^d}{\partial r} > 0$$

- Take logs of $\mathcal{A}^s(r^*, \eta) = \mathcal{A}^d(r^*, \eta)$ and differentiate:

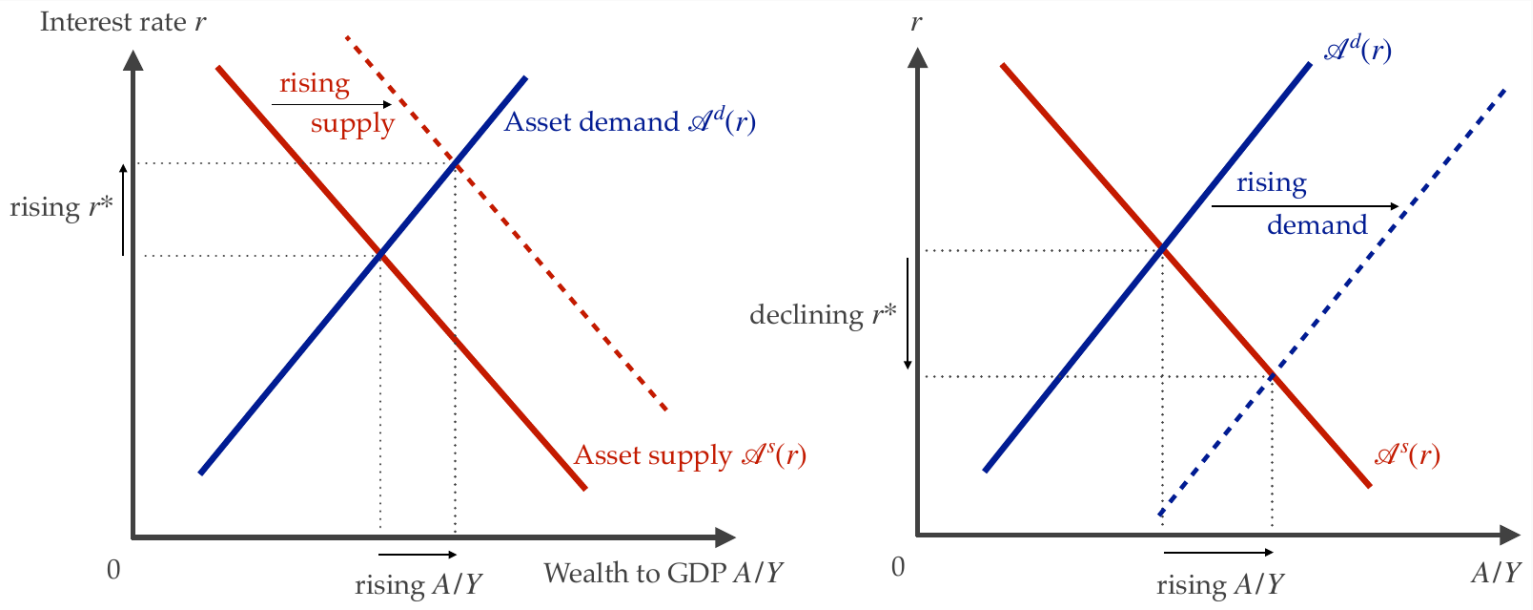
$$dr^* = \frac{\partial \log \mathcal{A}^s - \partial \log \mathcal{A}^d}{\epsilon^s + \epsilon^d}, \qquad d \log \frac{A}{Y} = \frac{\epsilon^d \, \partial \log \mathcal{A}^s + \epsilon^s \, \partial \log \mathcal{A}^d}{\epsilon^s + \epsilon^d}$$

  where $\partial \log \mathcal{A}$ is the shift of a curve at a given r
  - r^* falls if and only if demand shifts more than supply
- Invert: the observed dr and $d \log(A/Y)$ reveal the shifts

$$\partial \log \mathcal{A}^s = d \log \frac{A}{Y} + \epsilon^s dr, \qquad \partial \log \mathcal{A}^d = d \log \frac{A}{Y} - \epsilon^d dr$$

- Only two numbers needed: $\epsilon^s \approx 20$ and $\epsilon^d \approx 9$ (from the model below)

Who won the race?



- Asset supply shifts out: A/Y up and r^* up
- Asset demand shifts out: A/Y up and r^* down
- Data: A/Y up and r down $\Rightarrow$ asset demand won the race

Model: overview

- Long run only: compare balanced growth paths, no transitions
- **Firms**: Cobb-Douglas, markup μ, tax τ on income net of depreciation
- **Government**: debt B, consumption G, social security T, tax τ on all net income
- **Households**: overlapping generations (ages $j = 0, \ldots, J$) and permanent types θ
  - life-cycle consumption-saving problem, bequest motive, taxes and social security
- **Two assets**: safe bonds (r^f) and risky private assets (r^r)
  - risk premium $rp = r^r - r^f$ taken as given, households only care about the average return r
- Growth: $1 + g = (1 + n)(1 + \gamma)$, with population growth n and productivity growth γ

Model: asset supply

- Solve the firm's problem from the FOC: Cobb-Douglas, markup μ, tax τ on profits net of depreciation. Capital FOC: $\frac{\alpha}{\mu}\frac{Y}{K} = \frac{r^r}{1-\tau} + \delta$, profits are a share $1 - 1/\mu$ of output
- Asset supply in closed form, with $r^r = r + \bar{s}\, rp$ the return on risky assets:

$$\mathcal{A}^s(r) = \underbrace{\frac{\alpha}{\mu}\frac{1}{\frac{r^r}{1-\tau} + \delta}}_{K/Y\text{: capital}} + \underbrace{(1-\tau)(1+g)\frac{1-1/\mu}{r^r - g}}_{\Pi/Y\text{: value of profits and rents}} + \underbrace{\frac{B}{Y}}_{\text{gov. debt}}$$

  - Π/Y: profits $(1-1/\mu)Y$ grow at g and are discounted at r^r
  - B/Y: set by the government, does not respond to r
- Semi-elasticity:

$$\epsilon^s = \underbrace{\frac{K}{A}\frac{1}{r^r + (1-\tau)\delta}}_{\text{lower } r\text{: more capital}} + \underbrace{\frac{\Pi}{A}\frac{1}{r^r - g}}_{\text{lower } r\text{: rents worth more}} \approx 20$$

Model: shifters of asset supply

- **Public debt** B/Y: from the data. Going forward, aging raises deficits (pensions, health care)
- **Markups** μ: less capital, but more profits to capitalize. With $r^r - g$ small, the second effect wins
- **Capital share** α: technology (automation, AI), cheaper investment goods
- **Depreciation** δ: shift toward short-lived capital (equipment, software) raises the user cost, less K/Y
- **Taxes** τ: raise the user cost and lower after-tax profits
- **Growth** $1 + g = (1 + n)(1 + \gamma)$: exogenous, from the data. Slower growth lowers the value of rents
  - slower population growth n: the demographic effect on asset supply
- **Risk premium** rp: at a given r, a higher rp raises r^r: less capital, profits worth less

Model: households

- Overlapping generations, each solving a deterministic life-cycle problem:

$$\max_{\{\tilde{c}_j, a_{j+1}\}} \sum_{j=0}^{J} \Phi_j \tilde{\beta}_j \frac{\tilde{c}_j^{1-1/\sigma}}{1-1/\sigma} \quad \text{s.t.} \quad \tilde{c}_j + \phi_j a_{j+1} = y_j(\theta) + (1+r)a_j$$

  - ϕ_j: survival probability from j to $j+1$, $\Phi_j = \prod_{k<j} \phi_k$. Start with $a_0 = 0$
  - $\tilde{c}_j$: consumption, including end-of-life spending and bequests
  - $y_j(\theta) = (1-\tau)\, w\, \lambda(\theta)\, (\bar{e}_j + T\bar{t}_j)$: after-tax labor income and social security
- Two sources of heterogeneity:
  1. **Age** j: income and wealth follow the life cycle
  2. **Permanent income** $\lambda(\theta)$: scales the whole income profile

  No idiosyncratic income risk: the only risk is mortality
- Preferences $\tilde{\beta}_j$ are set to match the observed profiles of consumption and wealth, by age and by income

Model: households in the data

- **Wealth by age and type** is read from the data, not computed:
  - by age: 2016 profiles of net worth a_j and labor income we_j (SCF)
  - by type: θ is the percentile of labor income at ages 25-44, $\lambda(\theta)$ from DINA. In the PSID, log wealth on log labor income has a slope of 1.61, so $\bar{a}_j(\theta) = \bar{a}_j \lambda(\theta)^{0.61}$
- **Moments for** ϵ^d, from π_j, a_j and effective consumption $\tilde{c}_j$ (PSID consumption plus end-of-life spending):

$$A^h = \sum_j \pi_j a_j, \qquad \tilde{C} = \sum_j \pi_j \tilde{c}_j, \qquad \mathbb{E}Age_a = \frac{\sum_j \pi_j a_j j}{A^h}$$

$$\mathbb{E}Age_{\tilde{c}} = \frac{\sum_j \pi_j \tilde{c}_j j}{\tilde{C}}, \qquad \mathrm{Var}Age_{\tilde{c}} = \frac{\sum_j \pi_j \tilde{c}_j \left(j - \mathbb{E}Age_{\tilde{c}}\right)^2}{\tilde{C}}$$

- **Interpretation**:
  - $\mathbb{E}Age_a$: age of the owner of the average dollar of wealth
  - $\mathbb{E}Age_{\tilde{c}}$: age at which the average dollar is consumed
  - $\mathrm{Var}Age_{\tilde{c}}$: spread of consumption over the life cycle, the room to substitute

The slope of asset demand

Proposition 2 (with $r = g$)

$$\epsilon^d = \frac{1}{1+r}\Big(\underbrace{\sigma \frac{\tilde{C}}{(1+g)A^h}\mathrm{Var} Age_{\tilde{c}}}_{\text{substitution}} + \underbrace{\mathbb{E}Age_{\tilde{c}} - \mathbb{E}Age_a}_{\text{income}}\Big)$$

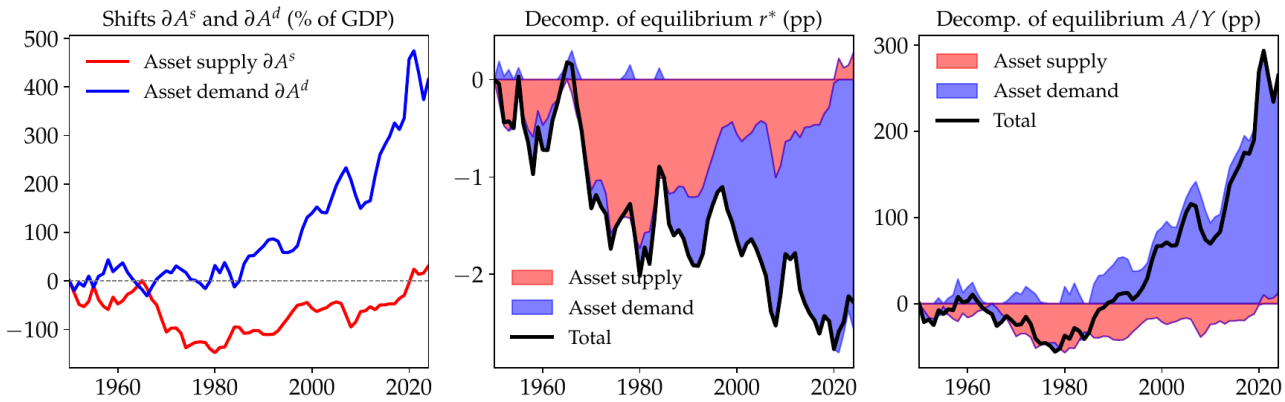
$Age_{\tilde{c}}$, Age_a: age of a random unit of consumption or of wealth in the population

- Depends only on the **age** structure of consumption and wealth, plus the EIS σ
- **Substitution** ($\sigma \times 50.2$): a higher r tilts consumption toward old age, so households save more when young
- **Income** (-7.4): wealth is about 7 years older than consumption. A higher return means less saving is needed for retirement
- With $\sigma = 1/3$, $\epsilon^d \approx 9$. Representative agent: $\epsilon^d = \infty$, so asset demand shifters cannot move r^*

Model: shifters of asset demand

- **Compositional**: change the weights on ages and types, not behavior. Reweight the 2016 data profiles
  - + **Demographics**: an older population puts more weight on asset-rich, low-income ages
  - + **Productivity growth**: slower growth across cohorts makes older cohorts relatively richer, like aging
  - + **Inequality**: income moves to the rich, who hold more wealth relative to their income, as in Straub (2019)
- **Life-cycle**: change saving behavior, measured with first-order formulas
  - − **Social security**: more generous pensions, less saving for retirement
  - **Longevity**: dropped, since most assets are not annuitized
- + **Foreigners**: $-NFA/Y$ taken from the data

Asset demand won the race



- Each year, two observations (dr, $d \log A/Y$ since 1950) and two unknown shifts:

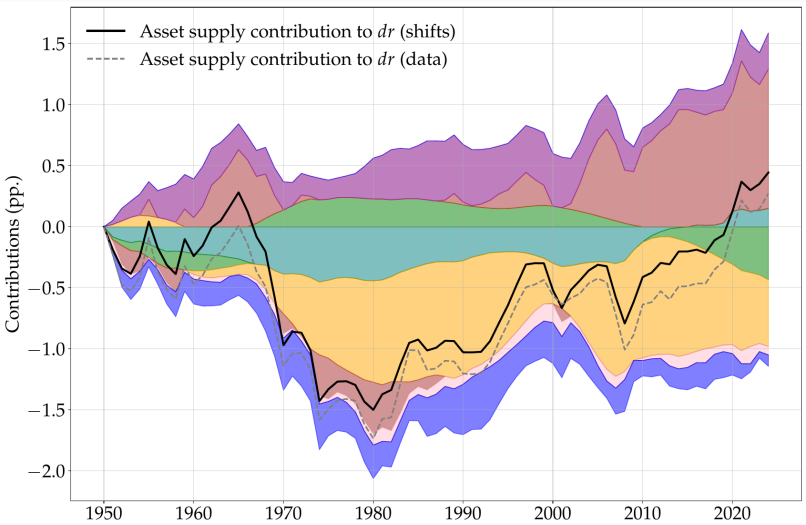
$$\partial \log \mathcal{A}^s = d \log \frac{A}{Y} + \epsilon^s dr, \qquad \partial \log \mathcal{A}^d = d \log \frac{A}{Y} - \epsilon^d dr$$

  change in A/Y = shift of the curve + move along it caused by dr
- r fell by about 2 pp. Along supply, A/Y rises by $\epsilon^s \times 0.02 \approx 40$ log points, most of the observed rise $\Rightarrow$ small supply shift
- Along demand, A/Y falls by $\epsilon^d \times 0.02 \approx 18$ log points, yet A/Y rose $\Rightarrow$ large demand shift

Source: AMRS (2025), Figure 4

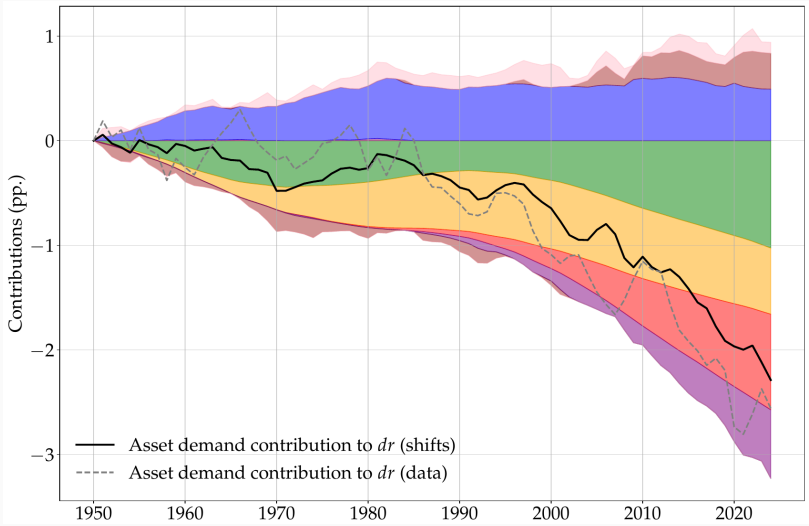
Which shifters moved r^*?

Asset supply



- Government debt
- Demographics
- Productivity growth
- Markups and rents
- Taxes
- Risk premium
- Capital intensity

Asset demand



- Demographics
- Productivity growth
- Inequality
- NFA
- Social security
- Labor share
- Taxes

Contribution of each shifter to $dr^* = \left(\partial \log \mathcal{A}^s - \partial \log \mathcal{A}^d\right) / (\epsilon^s + \epsilon^d)$, in pp.
Solid: sum of the shifters. Dashed: shift backed out from the data

Source: AMRS (2025), Figures 9(b) and 10(b)

The race behind us, 1950-2024

Asset supply

	% GDP	bp
Government debt	23	15
Demographics	−65	−43
Productivity growth	−82	−55
Markups and rents	170	114
Taxes	−10	−7
Risk premium	45	30
Capital intensity	−14	−9
Total explained	67	45
Total data	31	27

Asset demand

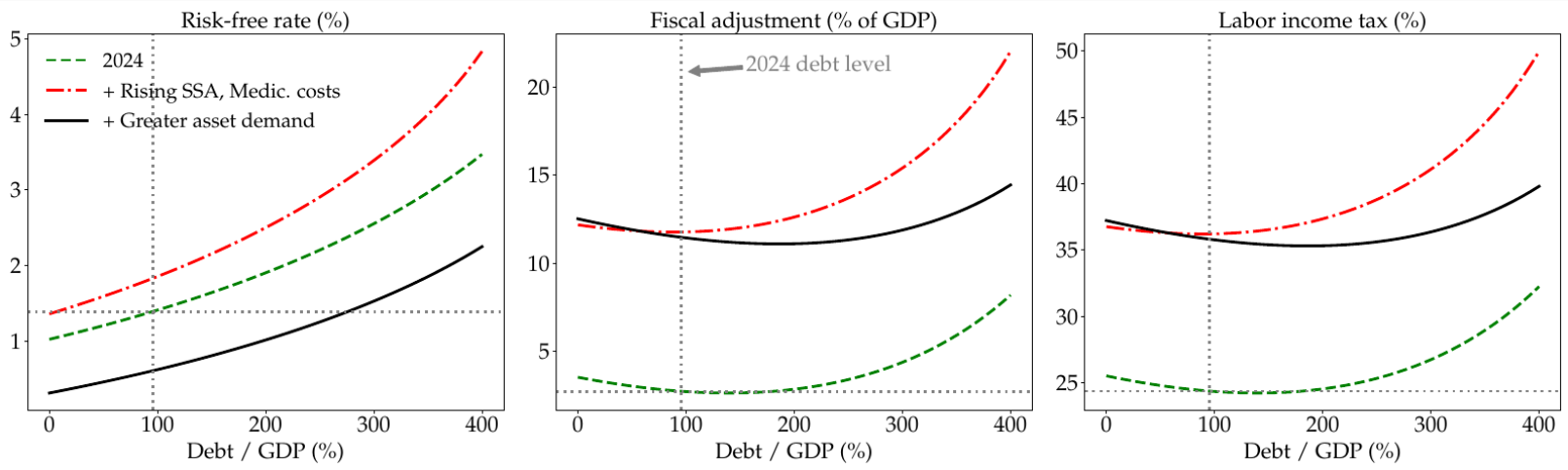
	% GDP	bp
Demographics	145	−103
Productivity growth	90	−63
Inequality	129	−91
NFA	79	−66
Social security	−70	49
Labor share	−48	34
Taxes	−15	11
Total explained	310	−229
Total data	415	−256

Shift of each curve (% of GDP) and contribution to dr^* (bp)

- Demographics and inequality: about −1 pp each. Slower growth and foreign inflows: about −0.7 pp each
- Markups and rents held r^* up by more than 1 pp: capitalized profits are an asset that households must hold

Source: AMRS (2025), Tables 1 and 2

The race ahead



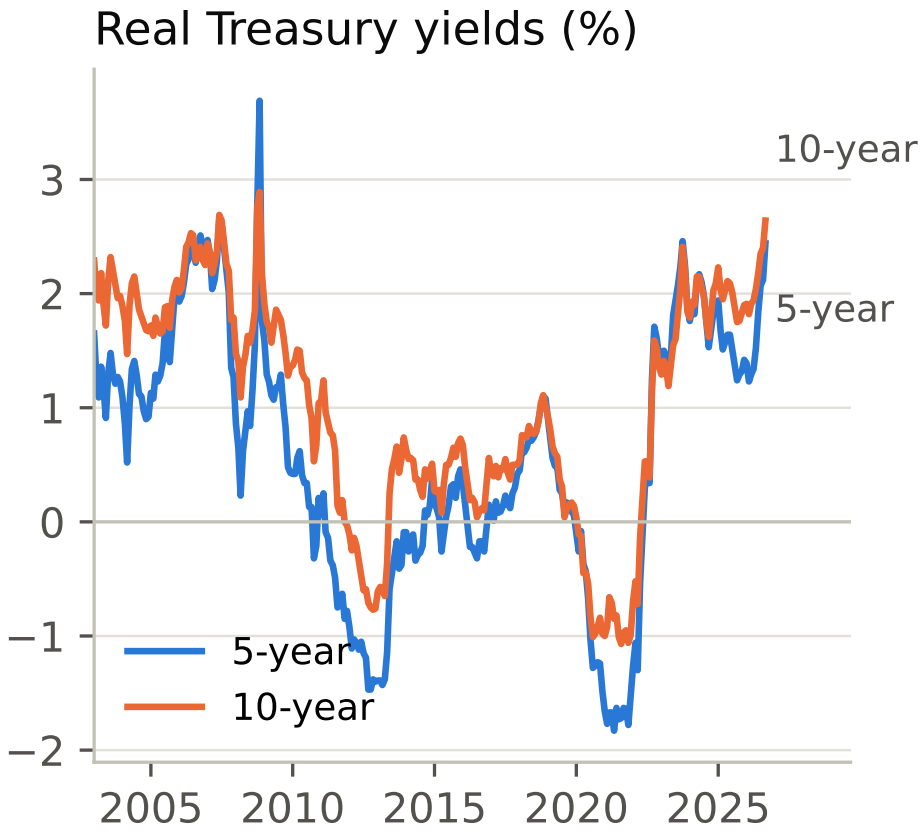
- Aging keeps raising asset demand: with fixed supply, r^* falls by 1.2 pp by 2100
- But old-age spending rises from 11% to 20% of GDP: without adjustment, debt grows by 500% of GDP and supply wins
- Debt stabilized in 2100 (figure): 250% of GDP at today's r, but it needs an adjustment of about 12% of GDP

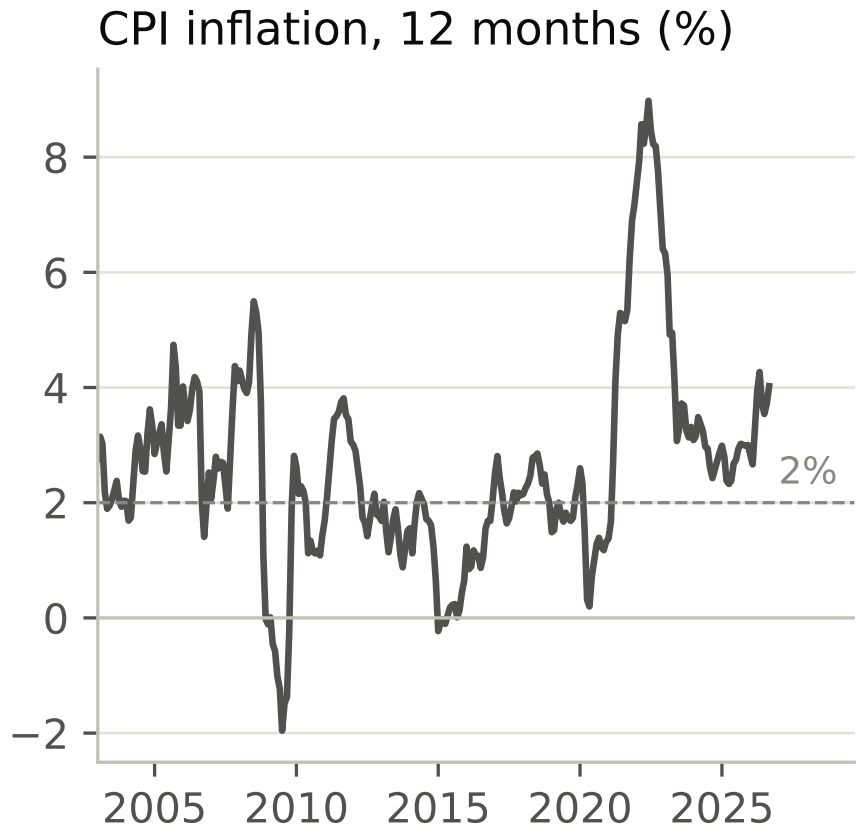
Source: AMRS (2025), Figure 15

Takeaways

- r^* is set by a race between asset supply (debt, capital, profits) and asset demand (households, foreigners)
- Heterogeneity makes asset demand imperfectly elastic, so aging and inequality move r^*
- 1950-2024: asset demand won by far (aging, inequality, slower growth, foreign inflows), and r^* fell
- Going forward: aging raises asset demand, but also deficits. There is room to borrow, but a large fiscal adjustment is unavoidable

Higher interest rates and inflation: a new normal?



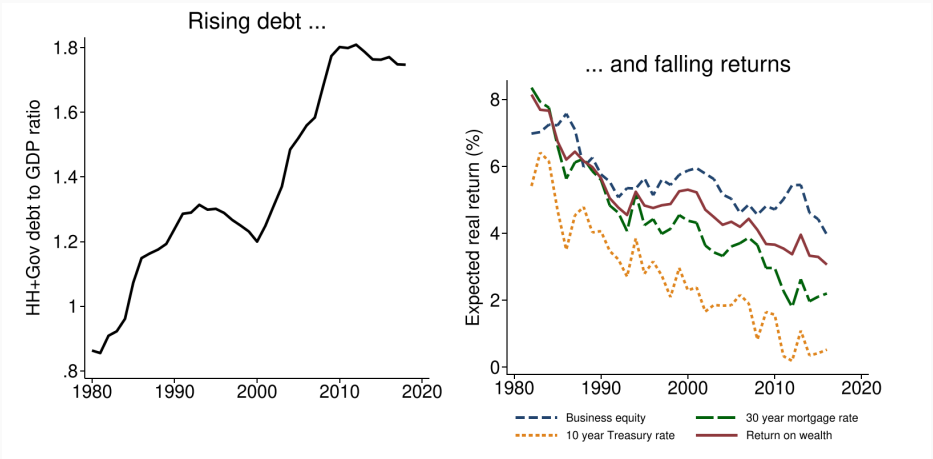


Source: FRED (FII5, FII10, CPIAUCSL), monthly.

Indebted demand
Mian, Straub and Sufi (2021)

Introduction

- Build on the non-homothetic preferences
- Highlights role of increasing household debt and connect to returns



- Argue that income inequality and less financial regulation leads to more borrowing and lower returns in GE
- **Note**: Would normally expect the opposite
  - If I want to borrow more, I will have to compensate savers by paying **higher** interest rate, not **lower**

Model

- Model
  - **Continuous** time
  - Endowment economy with $Y = 1$
- Savers s are unconstrained and maximize utility

$$\int_0^\infty e^{-\rho t} \log(c_t^s) + \upsilon(a_t^s) dt$$

- Borrowers b are constrained. They hold debt $d_t > 0$ up to a share ℓ of collateral $p_t = \int_t^\infty e^{-rt} \omega^b Y dt = \frac{\omega^b Y}{r}$

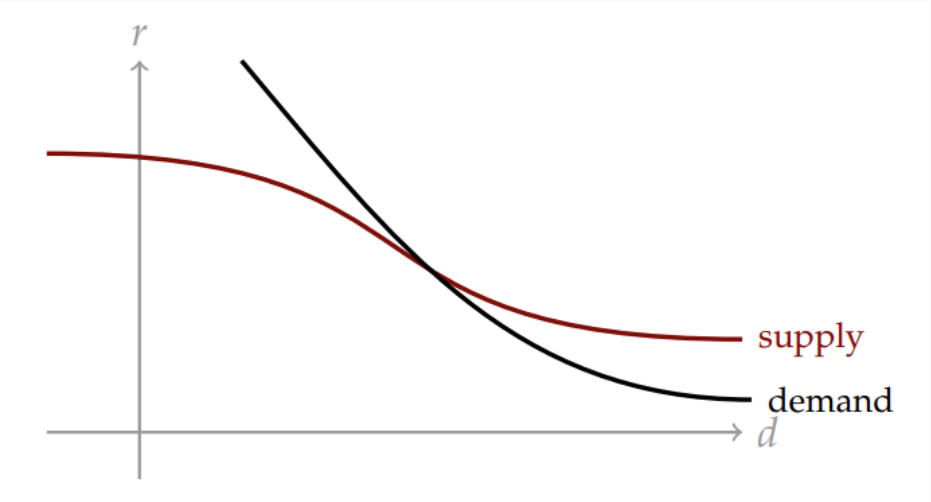
$$d_t = \frac{\ell \omega^b Y}{r}$$

- Equilibrium: debt owed by borrowing equal savings by savers:

$$d_t = a_t^s$$

Non-homothetic model

- Calibration
  - Interpret savers at top 1% of income distribution
  - Calibrate $\upsilon(a)$ such that savers have long-run MPC of 0.01
- **Non-homotheticity**: Large differences in SR across the two types
- Core insight of the paper: if savings is luxury good for rich, $\upsilon''(a_t^s) > 0$, saving schedule $a_t^s(r)$ is *downward sloping*



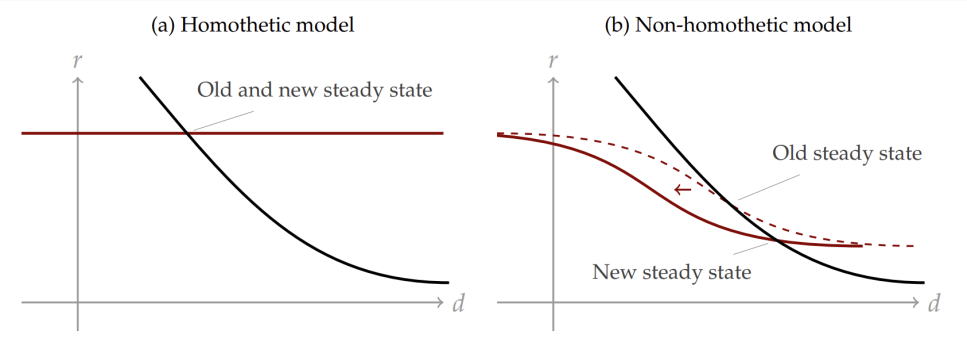
- Why? Assume rich agent consumes fixed amount $\bar{c}^s$ and saves the rest. Then $\bar{c}^s = r_t a_t^s \Leftrightarrow a_t^s = \frac{\bar{c}^s}{r_t}$
  - Higher r in the long run crowd out a_t^s in the long run

Results

- Increase in income inequality: increase in debt and lower interest rates
- Increase in financial deregulation $\rightarrow$ relax borrowing constraint: more borrowing and lower interest rates
  - Note: standard GE argument would imply higher rates
  - Lower rates arise due to downward sloping savings schedule of rich savers

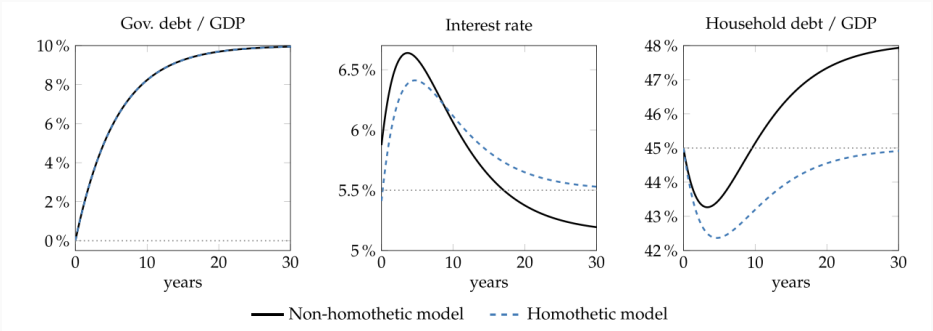
Results

- Increase in income inequality moves supply curve to the left
- Inequality increases $\rightarrow$ saving by rich increase $\rightarrow$ lower r $\rightarrow$ incentive to take more debt d



Results - Monetary and fiscal policy

- A permanent increase in public debt due to deficit spending shock
- Short run: r increase to induce more saving by rich savers to finance gov. debt
- Long run: the government pays rB to savers, financed by lower G or taxes on borrowers. Savers barely spend it, so demand falls and r must fall. Lower r crowds in household debt



- Opposite of AMRS, where more B raises r^*: there asset demand slopes up ($\epsilon^d > 0$), here the rich savers' demand slopes down
- For monetary policy: lower r means ZLB more likely, "less ammunition"

Exercise

Exercise

- Consider the following PE HA model:

$$V(a_{t-1}, z_t, p) = \max_{c_t} \frac{c_t^{1-\sigma}}{1-\sigma} + \phi_a \frac{a_t^{1-\sigma_a}}{1-\sigma_a} + \beta \mathbb{E}\left[V(a_t, z_{t+1}, p)\right]$$

$$\text{subject to}$$

$$c_t + a_t = a_{t-1}(1+r) + z_t p$$

$$a_t \geq 0$$

$$\ln z_t = \rho \ln z_{t-1} + \epsilon_t$$

- **Q1**: Derive the Euler equation of the model
- **Q2:** Update the EGM algorithm in *household_problem.py* with the new Euler
- **Q3:** Solve the model with 1) $\phi_a = 0$, 2) $\phi_a = 0.1, \sigma = \sigma_a = 1$, 3) $\phi_a = 0.1, \sigma = 1, \sigma_a = 0.7$
  - Compare the normalized policy function c/p
- **Q4:** Conduct an experiment where you permanently redistribute resources across households (change p). What are the aggregate effects across the models?

Summary

Summary and next week

- **Previously:** Long run wealth inequality
- **Today:** Explaining secular stagnation through HA models
  - Inequality and saving: non-homothetic preferences (Straub)
  - The race between asset supply and demand: aging, inequality, growth and debt (Auclert, Malmberg, Rognlie and Straub)
  - Indebted demand (Mian, Straub and Sufi)
- From now on: **Business cycles**
- Next lecture**: New Keynesian model + HANK**